

# Stochastic Process

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Md. Aminul Islam Shazid

# Outline

- ① Stochastic Process
- ② Markov Process

# Stochastic Process

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- A stochastic process  $\{X(t) : t \in T\}$  is a family of random variables indexed by a parameter  $t$ , which runs over an index set  $T$ .
- The parameter  $t$  usually denotes time.
- For any specific time  $t$ ,  $X(t)$  is a random variable.
- The index set  $T$  is called the parameter set.
- If  $T$  is countable,  $\{X(t)\}$  is a discrete time stochastic process. If  $T$  is an interval, finite or infinite,  $\{X(t)\}$  is a continuous time stochastic process.
- The set of possible values of  $X(t)$  at any time  $t$  is called the state space,  $S$ .

## Example

A business office has five telephone lines and any number of these lines may be in use at any time. The telephone lines are observed at regular intervals of 2 minutes.

$X$ : Number of lines in use in every 2 minutes.

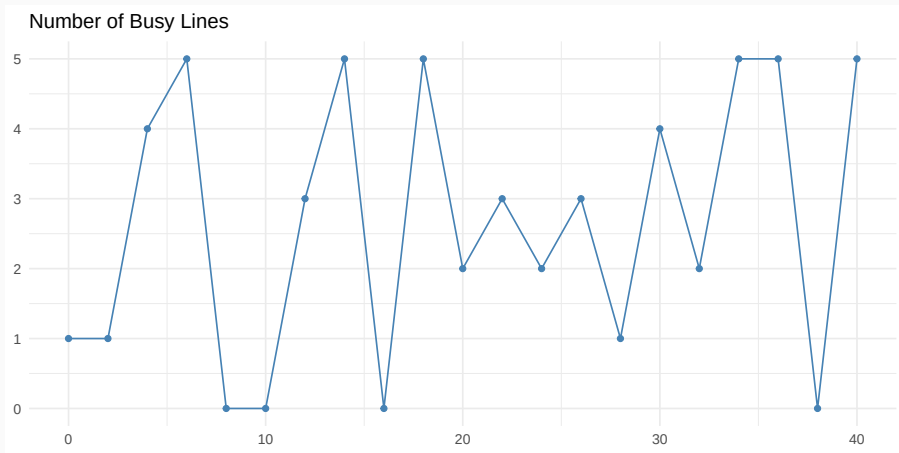
For  $T = 0, 1, 2, 3, \dots$ ,

$$X(t) : \{X(0), X(1), X(2), \dots\} \quad \text{e.g. } X(t) : \{3, 2, 5, 0, 1, 0, 3, \dots\}$$

Here,  $\{X(t) : t \in T\}$  is a stochastic process with parameter space  $T = \{0, 1, 2, 3, \dots\}$  and state space  $S = \{X : 0, 1, 2, 3, 4, 5\}$ .

# Graphical Example

An observation of this process might look like below.



## Examples of some stochastic processes

**1. Random walk model:** Let  $X_t$  be a random variable that can be any of the two states [up (+1) or down (−1)] at time  $t$ .

| $X_t$       | +1  | −1      |
|-------------|-----|---------|
| probability | $p$ | $1 - p$ |

Then, the process  $R_t = R_{t-1} + X_t$  is called a random walk model.

**2. Counting process:** A stochastic process  $\{N(t); t > 0\}$  is a counting process if  $N(t)$  represents the total number of events that have occurred in time  $t$ .

**3. Birth & death process:** A stochastic process  $\{N(t); t \geq 0\}$  with states  $\{n = 0, 1, 2, \dots\}$  for which transition from state  $n$  may go only to either of the states  $(n - 1)$  or  $(n + 1)$  is a birth and death process.

# Autocorrelation or Serial Correlation

- In statistics, the autocorrelation of a random process describes the correlation between values of the process at different times, as a function of the two times or of the time lag.
- Let  $X$  be some repeatable process, and  $i$  be some point in time after the start of that process. Then  $X_i$  is the value (or realization) produced by a given run of the process at time  $i$ .
- The autocorrelation between times  $s$  and  $t$  is defined as  $R(s, t) = \text{Corr}(X_t, X_s)$ , the correlation coefficient between  $X_t$  and  $X_s$ .



# Markov Process

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# Markov Process

- Consider a finite or countably infinite set of points  $(t_0, t_1, \dots, t_n, t)$ , where  $t_0 < t_1 < \dots < t_n < t$  and  $t, t_i \in T$  ( $i = 0, 1, 2, \dots, n$ );  $T$  is a parameter space of the process  $\{X(t)\}$ .
- The dependence exhibited by the process  $\{X(t) : t \in T\}$  is called Markovian dependence if the conditional distribution of  $X(t)$  for a given value of  $X(t_0), X(t_1), X(t_2), \dots, X(t_n)$  depends only on  $X(t_n)$ , which is the most recent known value of the process, i.e., if

$$\Pr[X(t) = x \mid X(t_n) = x_n, X(t_{n-1}) = x_{n-1}, \dots, X(t_0) = x_0] = \Pr[X(t) = x \mid X(t_n) = x_n]$$

The stochastic process exhibiting this property (Markov property) is called a Markov process.

# Types of Markov Process

| Parameter space |                                   | State space                            |            |
|-----------------|-----------------------------------|----------------------------------------|------------|
|                 |                                   | Discrete                               | Continuous |
| Discrete        | Markov chain                      | Discrete parameter, continuous state   |            |
| Continuous      | Continuous parameter, discrete MC | Continuous parameter, continuous state |            |

## Notations:

$p_i$  = Probability that the process is in state  $i$  at a given time

$p_{ij}^{(n)}$  =  $n$ -step transition probability of state  $j$  from state  $i$  =  $\Pr[\text{process moves from } i \text{ to } j \text{ in } n \text{ steps}]$

## Transition Probability Matrix (TPM)

A matrix containing probabilities of transition from one state to another. If there are  $k$  finite states of a Markov process, i.e.  $S = \{1, 2, \dots, k\}$ , the one-step TPM is

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1k} \\ p_{21} & p_{22} & \cdots & p_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1} & p_{k2} & \cdots & p_{kk} \end{pmatrix}$$

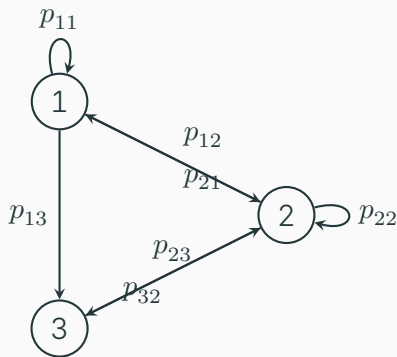
such that (i)  $p_{ij} \geq 0$ ,  $\forall i, j \in S$ ; (ii)  $\sum_j p_{ij} = 1$  (row total 1).

## Markov Process: Example 1

$X$  = living status. State space  $S = \{1, 2, 3\}$ ; where 1 = healthy, 2 = sick, 3 = dead.

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix}$$

Here,  $p_{11} > 0, p_{12} > 0, p_{13} > 0, p_{21} > 0, p_{22} > 0, p_{23} > 0, p_{31} = 0, p_{32} = 0, p_{33} = 1$ .



**Time-homogeneous Markov chains** (or stationary Markov chains) are processes where

$$\Pr[X_{t+1} = x \mid X_t = y] = \Pr[X_t = x \mid X_{t-1} = y]$$

for all  $t$ . The probability of the transition is independent of  $t$ .

# Classification of states

- **Accessible state:** A state  $j$  is said to be accessible from a state  $i$  (written  $i \rightarrow j$ ) if a system started in state  $i$  has a non-zero probability ( $p_{ij} > 0$ ) of transitioning into state  $j$  at some point.
- **Communicating states:** A state  $i$  is said to communicate with state  $j$  (written  $i \leftrightarrow j$ ) if both  $i \rightarrow j$  and  $j \rightarrow i$ . Here,  $p_{ij} > 0$  and  $p_{ji} > 0$ .
- **Absorbing state:** A state  $i$  is called absorbing if it is impossible to leave this state. Therefore, state  $i$  is absorbing if and only if  $p_{ii} = 1$  and  $p_{ij} = 0$  for  $i \neq j$ .

## Classification of states (cont.)

- **Transient state & recurrent state:** A state  $i$  is said to be transient if, given that we start in state  $i$ , there is a non-zero probability that we will never return to  $i$ . State  $i$  is recurrent (or persistent) if it is not transient. Recurrent states are guaranteed (with probability 1) to have a finite hitting time.
- **Periodic state & aperiodic state:** A state  $i$  has period  $k$  if any return to state  $i$  must occur in multiples of  $k$  time steps. A state is aperiodic if returns to state  $i$  can occur at irregular times ( $k = 1$ ).



## Multi-step transitions

- The  $n$ -step TPM is obtained by multiplying the one-step TPM by itself  $n$  times:  $P^{(n)} = P^n$ . Its  $(i, j)$ th entry,  $p_{ij}^{(n)}$ , is the probability of moving from state  $i$  to state  $j$  in  $n$  steps.
- If  $A = (p_1, p_2, \dots, p_k)$  is the probability vector (distribution over states) at time 0, then the distribution after  $n$  steps is

$$A^{(n)} = A \cdot P^n$$

- This can be computed either as  $A^{(n)} = A^{(n-1)} \cdot P$  (applying  $P$  one step at a time) or directly as  $A \cdot P^{(n)}$  once  $P^{(n)}$  is known — both give the same result.

## Example 2

There are two telephone lines in an office and any number of these lines may be in use at any given time. Telephone lines are observed at regular intervals of 2 minutes. The initial probability vector of the states is

$$A = (0.3, 0.5, 0.2)$$

and the one-step transition probability matrix is

$$P = \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

Assuming a time-homogeneous Markov chain  $X_t$ , determine the probabilities that no line, 1 line, and 2 lines are being used at (i)  $t = 1$ , (ii)  $t = 2$  (starting time  $t = 0$ ). Also find  $\Pr[X_0 = 1, X_1 = 0, X_2 = 2]$ .

Let 0: no line in use, 1: 1 line in use, 2: 2 lines in use, so  $S = \{0, 1, 2\}$ .

## Example — solution

**For  $t = 1$ :**

$$A^{(1)} = AP = (0.3, 0.5, 0.2) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} = (0.23, 0.51, 0.26)$$

**For  $t = 2$ :**

$$A^{(2)} = A^{(1)}P = (0.23, 0.51, 0.26) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} = (0.225, 0.497, 0.278)$$

## Example — solution

Find  $\Pr[X_0 = 1, X_1 = 0, X_2 = 2]$ .

$$\Pr[X_0 = 1, X_1 = 0, X_2 = 2] = \Pr[X_2 = 2 \mid X_1 = 0] \Pr[X_1 = 0 \mid X_0 = 1] \Pr[X_0 = 1] = p_{02} p_{10} p_{01}$$

using the Markov property,  $\Pr[X_2 = 2 \mid X_1 = 0, X_0 = 1] = \Pr[X_2 = 2 \mid X_1 = 0]$ .

So,

$$\Pr[X_0 = 1, X_1 = 0, X_2 = 2] = p_1 p_{10} p_{02} = 0.5 \times 0.3 \times 0.2 = 0.03$$

## Steady State Probabilities

If, for an irreducible Markov chain (only one class, so all states communicate), all states are aperiodic and positive recurrent (returns in finite time), the distribution  $A^{(n)} = A \cdot P^n$  converges as  $n \rightarrow \infty$ , and the limiting distribution is independent of the initial probabilities  $A$ :

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} a_j^{(n)} = p_j$$

These are called steady state probabilities, and satisfy

$$p_j > 0, \quad \sum_{j=0}^m p_j = 1, \quad p_j = \sum_{i=0}^m p_i p_{ij}, \quad j = 0, 1, \dots, m$$

Since there are  $m + 2$  equations but only  $m + 1$  unknowns, one equation is redundant — use  $m$  of the  $m + 1$  equations above together with  $\sum_j p_j = 1$ .

## Example 3

Find the steady state probabilities for the Markov chain in Example 2.

At steady state,

$$(p_0, p_1, p_2) = (p_0, p_1, p_2) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

which gives the system

$$-0.8p_0 + 0.3p_1 + 0.1p_2 = 0$$

$$0.6p_0 - 0.5p_1 + 0.4p_2 = 0$$

$$p_0 + p_1 + p_2 = 1$$

(the third original balance equation is redundant given  $\sum p_j = 1$ ).

## Example — solution

Solving the system by elimination gives

$$p_0 = \frac{17}{77}, \quad p_1 = \frac{38}{77}, \quad p_2 = \frac{22}{77}$$

These are the long-run (steady state) probabilities that 0, 1, and 2 telephone lines are in use, respectively, regardless of the initial distribution  $A$ .